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Fuel 83 (2004) 2427–2433

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XRD analysis of carbon stacking structure in coal during heat treatment

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Received 8 July 2002; accepted 2 June 2004

Available online 14 July 2004

Abstract

The fraction of the stacking structure (P_s) and the average number of aromatic layers (N_{ave}) in several raw coals were estimated by X-ray diffraction analysis (Standard Analysis of Coal by XRD: STAC-XRD). STAC-XRD is more appropriate for analyzing coal with low crystallinity than the classical XRD analysis method by using the interlayer spacing of aromatic layers (d_{002}) and the mean crystallite size along the *c* axis (L_c). STAC-XRD was also used to compare the development of the stacking structure in caking coal during heat treatment with that in non-caking coal. After heat treatment at 920 °C, N_{ave} of the caking coals was larger than that of the non-caking coals. The stacking of the aromatic layers in the caking coal temporarily developed at the fluidity stage at 400–440 °C, but in the 440–600 °C range, the release of volatile components partially disordered the stacked layers. The influence of heat-treatment conditions on carbon stacking structure in several kinds of coal was investigated by means of STAC-XRD. The heating rate required for the development of stacking structure in lower-rank coals was slower than that required for higher-rank coals.

Keywords: X-ray diffraction; Stacking structure; Caking coal; Non-caking coal; Heating rate

1. Introduction

X-ray diffraction (XRD) analysis is a fundamental method for evaluating carbon stacking structure. The degree of graphitization, the interlayer spacing (d_{002}), and the crystallite size (L_a, L_c) have been established as the parameters for evaluating the stacking structure of highly crystalline carbon materials [1,2]. The structure of coal has also been characterized by XRD, and the existence of crystallites in coal structure has been proven by the appearances of the peaks, corresponding to the 002, 100, and 110 reflections of graphite [3]. However, the vagueness of the XRD profile of coal with low crystallinity often makes quantitative evaluation of the stacking structure difficult. Forty to fifty years ago, Hirsch [4] and Diamond [5] proposed the statistical interpretation of XRD profiles for carbon materials with low crystallinity, but the recent development of computer techniques has enabled more-accurate calculations and detailed analysis of XRD profiles. In a previous study [6–8], we reported a standard analysis of carbon stacking structure in coal by XRD (Standard Analysis of Coal by XRD; STAC-XRD), and we used the method to analyze the stacking structure in coals and activated carbons. Fujimoto and Shiraishi [9] modified the Diamond's method for the estimation of carbon layer size. Sharma et al. [10] reported that good agreement was observed between the results from XRD analysis and the modified high-resolution transmission electron microscopy (HRTEM) technique for estimating the carbon stacking number and layer size in formaldehyde resin char.

The STAC-XRD method is based on the procedures proposed by Hirsch [4] and Shiraishi [11], and is established using the results from the actual analysis of several kinds of coals. The measurement conditions, and method to correct and analyze the XRD profiles are determined and simplified for the coals. Thus, STAC-XRD makes it possible to easily

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0016-2361/$ - see front matter
doi:10.1016/j.fuel.2004.06.019

estimate the parameters with respect to the stacking structure in coal and other carbon materials with low crystallinity. In a previous study [6–8], we analyzed the stacking structure of activated samples prepared from coal and coconut shell using STAC-XRD. We also used the method to clarify the mechanism of the degradation of the stacking structure in coal and coconut shell during activation. The coals were deashed and treated with hexane or methanol and then characterized by STAC-XRD. The solvent treatment partially destroyed the loose stacking structure; deashing had little effect.

In the present study, we used STAC-XRD to analyze the stacking structure in raw coals with carbon contents of 62–91 wt%, and we estimated the parameters with respect to the stacking structure. We also used STAC-XRD to investigate the changes in the stacking structures that occurred during heat treatment of caking and non-caking coals. In addition, we investigated the influence of the heat-treatment conditions on the development of the stacking structure.

2. Experimental

2.1. Sample preparation and heat treatment

The elemental analyses of the raw coals used in this study are summarized in Table 1. The coals were pulverized to particles of less than 149 μm in diameter and then dried in vacuo at 70 °C for 12 h. Foutuna (FO) and Buckskin (BS) are German and American brown coals, respectively; and Miike (MI), Ebenezer (EB), Newlands (NL), and Hebu Semi-Ant (HE) coals were obtained from the Japan Coal Energy Center. Pittsburgh #8 (PI) coal was selected from Argonne Premium Coal Samples. MI and PI are caking coals, and others are non-caking coals.

The coals were heat treated in a N_2 atmosphere at 350–920 °C for 1 h at a heating rate of 3 °C/min. To evaluate the influence of the heating rate, BS, EB, PI, and NL coals were pyrolyzed at 920 °C for 1 min at a heating rate of 300 °C/min in a N_2 atmosphere using an infrared image furnace (Shinku-Riko, RHL-E410).

2.2. XRD analysis

Prior to XRD analysis, the coal samples were stirred in 5 M HF for 24 h at ambient temperature to remove the peaks of mineral matters. The samples were then stirred in 5 M HCl for 24 h at ambient temperature and washed with pure water. In the previous study, we confirmed that the influence of the demineralization treatment on the stacking-structure parameters determined by the STAC-XRD was negligible [7].

In STAC-XRD, the fraction of the stacking structure (P_s), the distribution of the number of aromatic layers in the stacking structure (distribution of N), and the average number of N (N_{ave}) were estimated by analysis of the intensity of the 002 peak of carbon. P_s has been defined by Shiraishi [11] as the ratio of the number of carbon atoms in stacked structures to the number of carbon atoms in the entire sample. The distribution of N was calculated by a Fourier transform method [4]. The XRD profiles of the pulverized samples were measured using an X-ray diffractometer (Rigaku, Rotaflex RU-300) with Cu K_α radiation by a step-scanning method in the range of $2\theta =$ 10–41°. After the XRD profiles were corrected with adsorption, polarization, and atomic scattering factors, the parameters with respect to the stacking structure were estimated. Details of the STAC-XRD method have been reported previously [6].

The interlayer spacing of aromatic layers (d_{002}) in the samples was calculated from the peak position by means of Bragg's equation. The mean crystallite size along the c axis (L_c) was calculated from the width at the half maximum of the 002 peak using Scherrer's equation. The average number of N (N_c) was estimated from d_{002} and L_c by means of the following equation [10]:

$$N_c = \frac{L_c}{d_{002}}$$

Table 1
Elemental analyses of raw coals

Coal	Abbreviation	Wt%, daf					Ash (wt%)
		C	H	N	S	O[a]	
Foutuna[b]	FO	62.4	5.0	0.7	0.4	31.5	4.6
Buckskin[b]	BS	71.7	5.4	1.5	0.8	20.6	7.1
Miike[c]	MI	80.7	6.2	1.3	2.4	9.4	19.8
Ebenezer[b]	EB	81.2	6.1	1.6	0.6	10.5	14.9
Pittsburgh #8[c]	PI	82.0	5.6	1.2	0.9	10.3	8.4
Newlands[b]	NL	85.9	4.9	1.7	0.5	7.0	15.4
Hebu Semi-Ant[b]	HE	90.6	4.2	1.3	0.4	3.5	16.8

[a] Determined by difference.
[b] Non-caking coal.
[c] Caking coal.

3. Results and discussion

3.1. Carbon stacking structure in raw coals

Fig. 1 shows the XRD profiles of the demineralized raw coals. The peak at $2\theta = 20$–$30°$ corresponds to the 002 reflection of carbon due to the stacking structure of aromatic layers. The broadening of the 002 peak for the brown coals can be attributed to the small dimensions of the crystallites perpendicular to the aromatic layers. The 002 peaks of the higher-rank coals are sharper than those of the brown coals. Furthermore, the peaks shift to a higher 2θ value as the carbon content of the raw coals increases. Table 2 shows the d_{002}, L_c, and N_c values for the raw coals. The parameter d_{002} is a measure of the perfection in the stacking structure periodicity. The decrease in d_{002} with increasing carbon content is related to the development of the stacking structure (d_{002} for the graphite structure is 0.335 nm). However, assessing the stacking structure using only d_{002} is difficult. The L_c value indicates the thickness of the stacking structure along the c axis. The L_c and N_c values of the higher-rank coals were larger than those of the lower-rank coal.

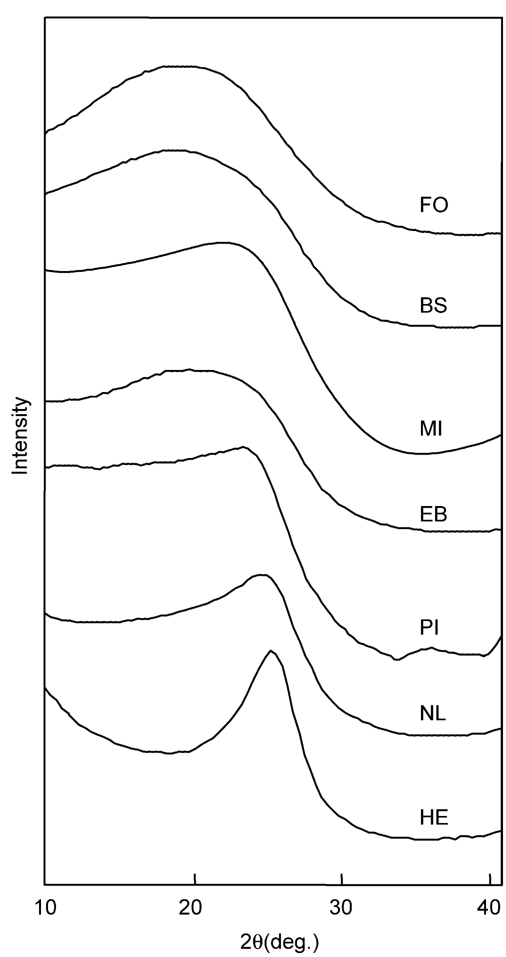

Fig. 1. XRD profiles of raw coals.

Table 2
Interlayer spacing (d_{002}), mean crystallite size (L_c), and average number of N (N_c) for the raw coals

Coal	d_{002} (nm)	L_c (nm)	N_c
FO	0.368	0.79	2.1
BS	0.356	0.71	2.0
MI	0.359	0.86	2.4
EB	0.365	0.92	2.5
PI	0.365	1.06	2.9
NL	0.351	1.25	3.6
HE	0.348	1.89	5.4

The P_s values increased linearly with increasing carbon content (Fig. 2). Fig. 3 shows the distribution of the number (N) of aromatic layers in raw coals. The distributions of N for the FO and BS coals did not differ significantly: in both of these coals, more than 70% of the stacking structure consisted of two layers. In the NL and HE coals, the proportion of two-layer structure decreased, and that of the three to five-layer structure increased, relative to the values for the FO, BS, MI, and EB coals. Fig. 4 shows the N_{ave} values for the raw coals calculated from the distribution of N. The value for the lower-rank coals with carbon contents of less than 72 wt% was approximately 2.2. For coals with a carbon content more than 81 wt%, the N_{ave} values increased with increasing carbon content. This pattern indicates that the stacking structure of the aromatic layers in the higher-rank coals (carbon content >81 wt%) differs from that in the brown coals.

The N_{ave} values (Fig. 4) of the higher-rank coals differed from the N_c values, whereas the variations in N_{ave} and N_c with carbon content were in agreement. In STAC-XRD, N_{ave} and the distribution of N are estimated by means of a Patterson function, $P(u)$, which represents the probability of finding a layer at a distance of u normal to a given layer, assuming that there is a distribution of crystallite sizes in the sample. The N_c value was estimated from d_{002} and L_c, which was calculated with Scherrer's equation. This equation is based on the assumption that the width of the diffraction line

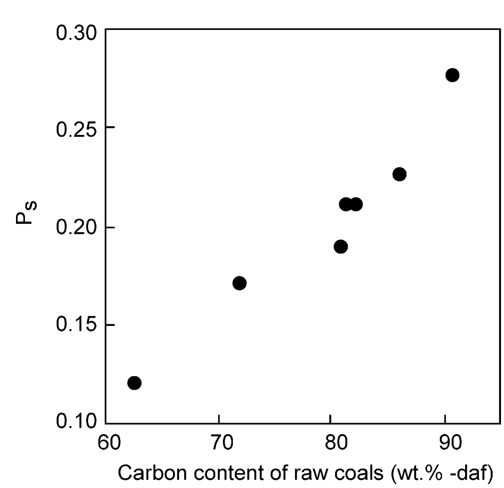

Fig. 2. P_s values of raw coals.

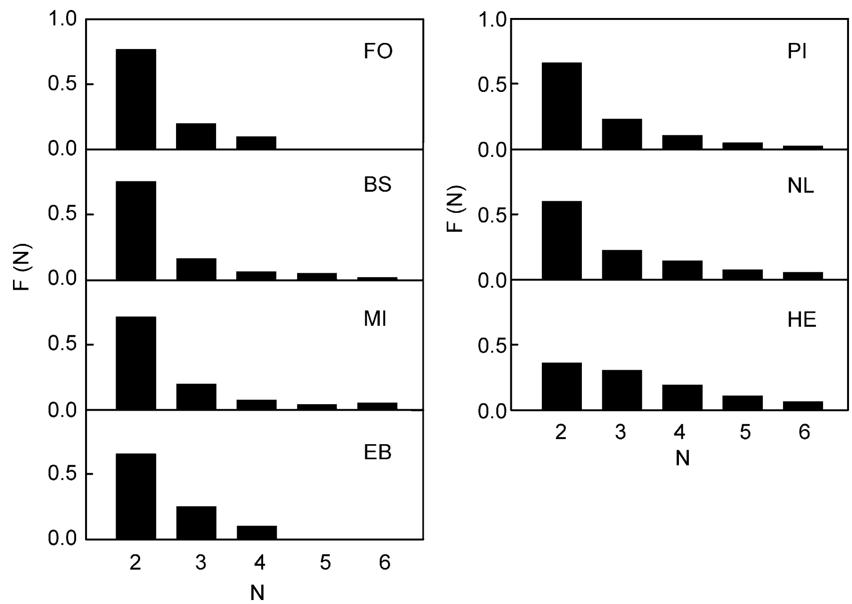

Fig. 3. Distribution of the number (N) of aromatic layers in raw coals.

depends only on the homogeneous size of crystallite without irregularity. In addition, the vagueness of d_{002} due to the broadness of the 002 peak in the XRD profile of coal makes the N_c value inaccurate. These facts indicate that N_{ave} is more appropriate than N_c for coal (that is, for carbon materials with low crystallinity and a distribution of crystallite sizes). The STAC-XRD method gives the distribution of N and P_s as well as N_{ave}. This is also an advantage over the classical XRD analysis method using d_{002} and L_c.

3.2. Change of stacking structure in caking coal during heat treatment

Table 3 shows solid yields (daf) when caking (PI, MI) and non-caking (EB) coals were heat treated at various temperatures. In all three samples, the decrease in the yield after heat treatment at 350 °C was 7–9 wt%, and was mainly due to the formation of gaseous products such as H_2O and CO_2. Most of the volatile components were released in the 350–600 °C range [12].

Fig. 5 shows the P_s values of caking (PI, MI) and non-caking coals (EB) as a function of heat-treatment temperature. The P_s values remained unchanged in the 350–600 °C range and then increased with increasing heat-treatment temperature above 600 °C. The P_s value of PI coal at 920 °C was higher than the values for MI and EB coals.

Fig. 6 shows the N_{ave} values of caking (PI, MI) and non-caking coals (EB) as a function of heat-treatment temperature. Although carbon content, P_s, and N_{ave} values for these caking coals without heat treatment were approximately equal to those for the non-caking coal (Table 1 and Figs. 2 and 4), the temperature dependence of N_{ave} for the caking coals differed remarkably from that of the non-caking coal. The N_{ave} value of unheated PI coal was 2.343, but the value increased to 2.672 after heat treatment at 400 °C, and to 2.803 at 440 °C. The N_{ave} value for MI coal also increased after heat treatment at 400 °C. After heat treatment at 920 °C, the N_{ave} values of the caking coals were larger than the value for the non-caking coal. The N_{ave} values of caking and non-caking coals after heat treatment at 427, 800, and 1200 °C in a helium atmosphere were analyzed by Sharma et al., using the modified HRTEM technique [13]. Their results agreed with ours: that is, N_{ave} for caking coal increased significantly at 427 °C and was larger than that of non-caking coal after heat treatment at 800 and 1200 °C.

Table 4 shows the thermoplastic properties of the caking coals estimated from a temperature-variable dynamic viscoelastic measurement. The method represents coal plasticity as well as Gieseler plastometry [14]. Table 3 shows the distribution of N and d_{002} for caking (PI, MI) and

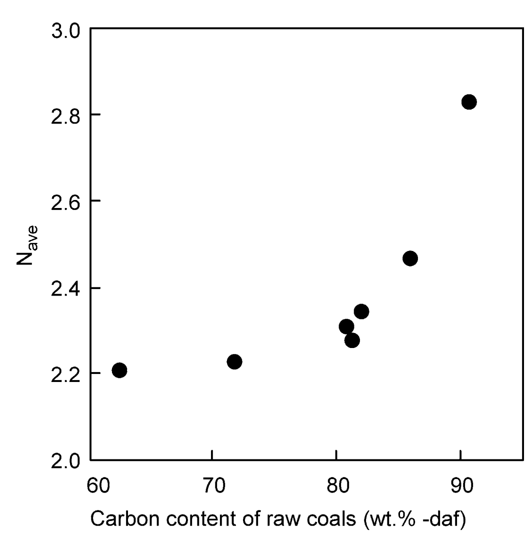

Fig. 4. N_{ave} values of raw coals.

Table 3
Yields and distribution of N, and d_{002} for caking (PI, MI) and non-caking coals (EB)

Coal	H-T[a] (°C)	Yield (wt%, daf)	Distribution of N (%)					d_{002} (nm)
			2	3	4	5	6	
PI	350	90.6	63.9	20.2	9.6	3.8	1.2	3.59
	400	76.1	41.2	28.4	17.2	9.1	0.9	3.51
	440	66.8	34.2	32.1	19.5	8.8	2.5	3.51
	600	59.6	40.0	35.9	15.3	3.2	3.2	3.48
	760	57.9	39.4	37.0	16.4	4.6	2.6	3.56
	920	57.1	37.9	37.4	16.0	5.1	1.0	3.51
MI	350	91.1	63.5	24.2	8.9	0.1	0.2	3.54
	400	74.5	35.6	37.5	17.2	5.0	1.0	3.51
	440	69.4	30.9	47.0	18.4	2.5	1.0	3.48
	600	63.3	43.0	40.5	12.4	3.6	0.3	3.48
	760	56.3	40.3	40.4	15.9	3.4	0.3	3.56
	920	54.9	48.6	33.8	11.6	4.4	0.1	3.56
EB	350	93.2	61.3	28.5	7.4	1.9	0.8	3.68
	400	81.7	50.1	28.7	13.7	1.1	0.6	3.53
	440	73.1	50.9	32.6	13.9	3.6	1.3	3.53
	600	67.1	55.1	25.7	14.5	4.0	0.0	3.56
	760	64.0	53.0	31.2	12.4	2.6	0.5	3.56
	920	62.7	51.7	27.3	15.3	3.2	1.7	3.56

[a] Heat-treatment temperature.

non-caking (EB) coals after heat treatment at the temperatures indicated. After heat treatment at 400 and 440 °C, the percentage of two-layer structure in the caking coals decreased greatly, and that of three to six-layer structure increased. The d_{002} value, however, did not show a definite dependence upon heat-treatment temperature.

Previous studies with respect to the mechanism of the thermoplasticity of caking coal have proposed that lower-molecular-weight components in caking coal, which are present in the raw coal or are generated by cleavage of covalent bonds during pyrolysis, play an important role in the softening and fluidity [14–17]. For heat-treated coals obtained at the fluidity stage, the maximum Gieseler fluidity has been reported to increase as the yield of solvent-soluble fraction increases [18]. Since the solvent-soluble fraction corresponds to the lower-molecular-weight components, the increase in the maximum Gieseler fluidity of caking coal indicates that lower-molecular-weight components promote the mobility of coal structure at the fluidity stage. The increase in the mobility of the structure in caking coal at the fluidity stage has also been analyzed by proton magnetic resonance thermal analysis and by an NMR microimaging method [19,20]. In this study, therefore, the increase in N_{ave} values for the caking coals at around the maximum fluidity temperature, 420 and 427 °C for MI and PI coals, respectively, can be explained by the fact that the development of the stacking structure of the aromatic layers was accelerated by the increase in the mobility of the structure at the fluidity stage. On the other hand, as shown in Fig. 5, the change of P_s of the caking coals at the fluidity stage was small. This fact suggests that the stacking

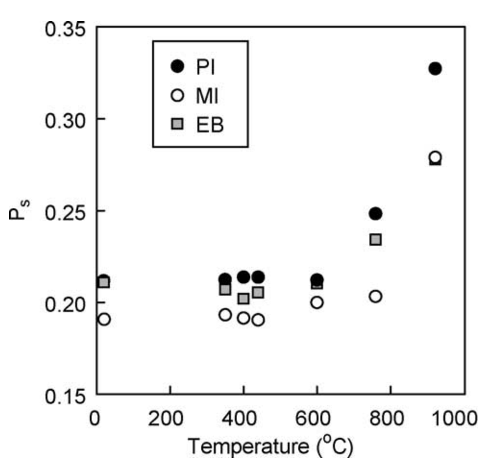

Fig. 5. P_s values of caking (PI, MI) and non-caking (EB) coals as a function of heat-treatment temperature.

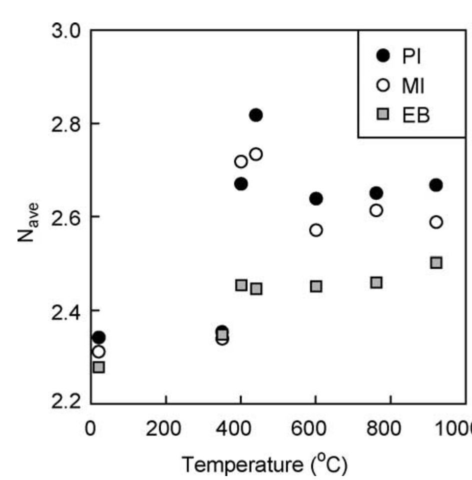

Fig. 6. N_{ave} values of caking (PI, MI) and non-caking (EB) coals as a function of heat-treatment temperature.

Table 4
Thermoplastic properties of caking coals

Coal	Softening temperature (°C)	Maximum fluidity temperature (°C)	Resolidification temperature (°C)
MI	355	420	460
PI	369	427	483

structure of the aromatic layers was partially developed at the fluidity stage but that there was no change in the fraction of the stacked structure relative to the whole structure. That is to say, the number of aromatic layers in the stacking structure increased, and the aromatic layers without stacking in the raw coal did not contribute to the increase in N_{ave} at 400–440 °C.

The N_{ave} value of the caking coals after heat treatment at 600 °C was smaller than that at 400 and 440 °C. We have previously reported that most of the volatile matters corresponding to the lower-molecular-weight components are released from the coal structure in the 440–600 °C range [12]. Thus, the decrease in N_{ave} of the caking coals at 600 °C may be interpreted by taking into account the fact that the stacking of aromatic layers developed at the fluidity stage was partially destroyed by the release of the lower-molecular-weight components incorporated in the stacking structure [21].

3.3. Influence of heat-treatment conditions on carbon stacking structure

In order to investigate the influence of the heating rate at 920 °C on the development of the stacking structure in coal, we compared the P_s and N_{ave} values after heat treatment for 1 min at a heating rate of 300 °C/min (fast treatment) with the values after heat treatment for 1 h at a rate of 3 °C/min (slow treatment). Table 5 shows yields and elemental analyses of BS, EB, PI, and NL coals after fast and slow treatments at 920 °C. The (O+S) value of BS coal after fast treatment was 8.2 wt%, which was larger than that after slow treatment. Since the sulfur content of BS coal is much lower than the oxygen content (Table 1), the (O+S) value can be regarded as the oxygen content. Fig. 7 shows the influence of heating rate at 920 °C on the P_s and N_{ave} values of coals. Both the P_s and N_{ave} value of BS coal after fast treatment were smaller than those after slow treatment. In the case of other coals,

Table 5
Yields and elemental analyses of coals after heat treatment at 920 °C

Coal	Reaction condition		Yield (wt%, daf)	Wt%, daf			
	Time	Heating rate (°C/min)		C	H	N	(O+S)[a]
BS	1 min	300	62.2	89.5	1.4	0.9	8.2
	1 h	3	61.8	94.8	1.0	0.8	3.4
EB	1 min	300	58.8	93.5	1.3	0.9	4.3
	1 h	3	57.1	95.2	1.0	1.2	2.3
PI	1 min	300	58.3	95.2	1.2	1.2	2.4
	1 h	3	62.7	93.5	0.9	1.4	4.2
NL	1 min	300	64.1	93.7	1.4	1.3	3.6
	1 h	3	64.7	95.8	1.1	1.3	1.8

[a] Determined by difference.

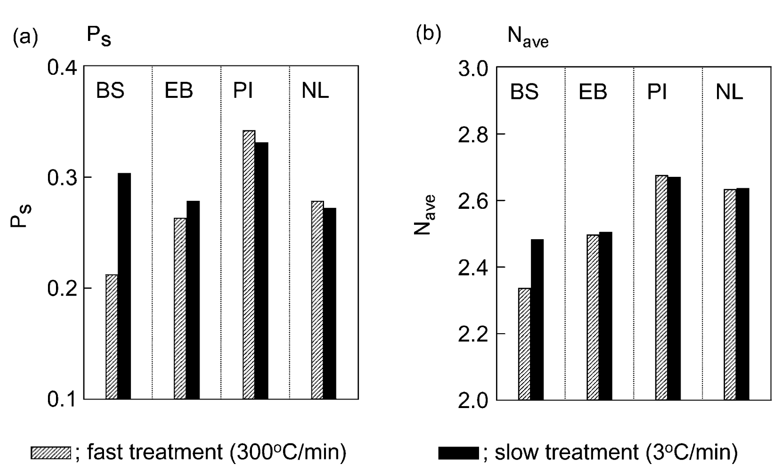

Fig. 7. Influence of heating rate at 920 °C on P_s and N_{ave} values of BS, EB, PI, and NL coals.

the P_s and N_{ave} differences between fast and slow treatment was small. BS coal, which is a lower-rank coal, contains smaller hexagonal carbon layers and a higher ratio of the bridged structures such as methylene and ether bonds [22]. Thus, this suggests that BS coal requires a slower heating rate to develop stacking structure than higher-rank coal does.

4. Conclusions

(1) The P_s, N_{ave}, and distribution of N values for several raw coals were estimated by STAC-XRD. Furthermore, the parameters with respect to the stacking structure, d_{002}, L_c, and N_c, calculated by the classical XRD analysis method were compared with those estimated by STAC-XRD. We concluded that STAC-XRD was more appropriate for the analysis of coal with low crystallinity.

(2) The changes in the stacking structures of caking and non-caking coals during heat treatment were investigated by STAC-XRD. Although the carbon content, P_s, and N_{ave} values of caking (PI, MI) coals without heat treatment were approximately equal to those of the non-caking (EB) coal used in this study, the temperature dependence of the change in N_{ave} values of these caking coals differed remarkably from the temperature dependence of the non-caking coal. The stacking structure of the aromatic layers was partially developed at the fluidity stage, but the fraction of the stacked structure to the whole structure at 400–440 °C did not change. Further, we determined that the stacking of aromatic layers that developed at the fluidity stage was partially destroyed by the release of the lower-molecular-weight components incorporated in the stacking structure.

(3) The P_s and N_{ave} values after fast treatment at 920 °C were compared with the values after slow treatment. The P_s and N_{ave} values of BS coal after fast treatment were smaller than those after slow treatment. By contrast, the difference of P_s and N_{ave} values of other coals between the fast and slow treatments was small. This result indicates that the lower-rank coal requires a slower heating rate to develop stacking structure than higher-rank coal does.

Acknowledgements

This study was supported by Coal Utilization Fundamental Technology Development, sponsored by the Center for Coal Utilization, Japan (CCUJ). We are deeply grateful to Dr M. Shiraishi (Tokai University) and Dr T. Yoshida (Forestry and Forest Products Research Institute) for useful discussions.

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